Notes: Imbens & Angrist (Econometrica, 1994)
Identification and Estimation of Local Average Treatment Effects

Contents

1 Big picture

- **Question.** What causal effect is identified by instrumental variables when treatment effects are heterogeneous and treatment take-up is imperfect?
- **Main answer.** Under an exclusion-type instrument condition and a monotonicity condition, IV identifies a **local average treatment effect** (LATE), not necessarily the average treatment effect for everyone or for all treated units.
- **Core contribution.** The paper formalizes the population whose causal effect IV identifies: individuals whose treatment status is changed by the instrument.
- **Key population.** These individuals are now called **compliers**: those who receive treatment under one value of the instrument but not under another.
- **Main identification formula.** For a binary instrument, the Wald ratio equals the average treatment effect for compliers.
- **Why this matters.** If treatment effects differ across individuals, IV is not a generic treatment effect. It is a weighted average of causal effects for instrument-induced movers.
- **Monotonicity.** The key extra restriction rules out people whose treatment status moves opposite to the instrument. In later language, it rules out **defiers**.
- **Multi-valued IV.** With a discrete multi-valued instrument and a scalar instrumenting function $g(Z)$, the IV estimand is a weighted average of adjacent LATEs when $g(Z)$ moves monotonically with the propensity score $P(Z)$.
- **Practical implication.** Empirical papers using natural experiments should describe the compliers: the effect is for people induced by the instrument, not necessarily the whole population.
- **Economic takeaway.** The paper turns IV from a mechanical estimator into a causal estimand with a transparent population interpretation.

2 Data and measurement objects

This is a theoretical econometrics paper, so there is no empirical dataset. The paper's key objects are potential outcomes, potential treatment decisions, and instruments.

2.1 Potential outcomes

Each individual i has two potential outcomes:

$$Y_i(0) = \text{outcome without treatment}, \qquad Y_i(1) = \text{outcome with treatment}.$$

The treatment indicator is $D_i \in \{0,1\}$, and the observed outcome is

$$Y_i = Y_i(D_i) = D_iY_i(1) + (1 - D_i)Y_i(0).$$

The individual treatment effect is

$$Y_i(1) - Y_i(0),$$

but this is never observed directly because only one potential outcome is realized.

Interpretation: The paper follows the potential-outcomes tradition rather than defining causality as a regression coefficient. This is essential because LATE is a causal average over a specific latent subpopulation.

2.2 Potential treatment status under instruments

Let Z_i be an instrument and let $\mathcal{Z}$ be its support. For each possible instrument value z, define

$$D_i(z) = \text{treatment status individual } i \text{ would have if } Z_i = z.$$

The realized treatment is

$$D_i = D_i(Z_i).$$

Interpretation: The notation $D_i(z)$ is the treatment-side analog of potential outcomes. It lets the paper define who is moved by the instrument without observing every counterfactual treatment status.

2.3 Propensity score induced by the instrument

Define

$$P(z) = \mathrm{E}[D_i \mid Z_i = z].$$

For a binary instrument, $P(1) - P(0)$ is the first-stage effect. For a multi-valued instrument, $P(z)$ describes how the treatment probability varies across instrument values.

Interpretation: A useful instrument must shift treatment probabilities. If $P(z)$ is constant, there is no first stage and no IV estimand.

2.4 Complier groups

For two instrument values z and w, the relevant mover group is

$$\{i : D_i(z) - D_i(w) = 1\}.$$

The associated local average treatment effect is

$$\alpha_{z,w} = \mathrm{E}\left[Y_i(1) - Y_i(0) \mid D_i(z) - D_i(w) = 1\right].$$

Interpretation: The target group depends on the instrument contrast. Different instruments, or different margins of the same instrument, can identify different causal averages.

3 Models and theoretical specifications

3.1 Condition 1: existence of instruments

The instrument condition has two pieces:

1. For all $w \in \mathcal{Z}$, the triple $(Y_i(0), Y_i(1), D_i(w))$ is jointly independent of Z_i.
2. $P(w) = \mathrm{E}[D_i \mid Z_i = w]$ is a nontrivial function of w.

Interpretation: The first part is the exclusion/exogeneity requirement. It says the instrument is as good as random with respect to potential outcomes and potential treatment decisions. The second part is relevance: the instrument must affect treatment probability.

3.2 Why Condition 1 alone is not enough

For two values z and w with $P(z) > P(w)$, the difference in conditional mean outcomes can be decomposed as

$$\mathrm{E}[Y_i \mid Z_i = z] - \mathrm{E}[Y_i \mid Z_i = w] = \mathrm{E}\left[(D_i(z) - D_i(w))(Y_i(1) - Y_i(0))\right].$$

If some people move from 0 to 1 and others move from 1 to 0, the two groups' causal effects can offset each other.

Interpretation: IV may fail to identify a meaningful positive average causal effect even if every individual has a positive treatment effect. The problem is not the exclusion restriction alone; it is the possibility of opposite-direction movers.

3.3 Condition 2: monotonicity

For all $z, w \in \mathcal{Z}$,

$$D_i(z) \geq D_i(w) \quad \text{for all } i,$$

or

$$D_i(z) \leq D_i(w) \quad \text{for all } i.$$

Interpretation: Monotonicity means the instrument moves treatment in one direction for everyone. In later terminology, this rules out defiers. It is not testable in general and must be argued from institutional details.

3.4 Theorem 1: binary-instrument LATE identification

If Conditions 1 and 2 hold, then for any z, w with $P(z) \neq P(w)$,

$$\alpha_{z,w} = \mathrm{E}\left[Y_i(1) - Y_i(0) \mid D_i(z) - D_i(w) = 1\right]$$

is identified by

$$\frac{\mathrm{E}[Y_i \mid Z_i = z] - \mathrm{E}[Y_i \mid Z_i = w]}{\mathrm{E}[D_i \mid Z_i = z] - \mathrm{E}[D_i \mid Z_i = w]}.$$

Interpretation: The numerator is the reduced form effect of the instrument on the outcome. The denominator is the first stage. Their ratio is the average causal effect for those moved by the instrument.

3.5 Condition 3: valid scalar functions of multi-valued instruments

For a scalar function $g(Z)$, the paper requires $g(z)$ to be monotone in $P(z)$, and

$$\mathrm{Cov}(D, g(Z)) \neq 0.$$

Interpretation: For multi-valued instruments, the choice of instrumenting function matters. If $g(Z)$ orders instrument values consistently with their treatment propensities, the resulting IV estimand has a clean nonnegative-weight LATE interpretation.

3.6 Theorem 2: IV as weighted average of LATEs

Let Z have support $\{z_0, z_1, \ldots, z_K\}$ ordered so that $P(z_k)$ increases in k. Then

$$\alpha_g^{IV} = \frac{\mathrm{Cov}(Y, g(Z))}{\mathrm{Cov}(D, g(Z))} = \sum_{k=1}^{K} \lambda_k \alpha_{z_k, z_{k-1}},$$

where the weights λ_k are nonnegative and sum to one.

A compact version of the weights is

$$\lambda_k \propto \big(P(z_k) - P(z_{k-1})\big) \sum_{l=k}^{K} \pi_l \left(g(z_l) - \mathrm{E}[g(Z)]\right),$$

where $\pi_l = \Pr(Z = z_l)$.

Interpretation: IV with a multi-valued instrument does not generally identify one single margin unless the instrument is binary. It averages adjacent complier effects, with weights determined by first-stage increments and the instrumenting function.

3.7 Theorem 3: asymptotic distribution of IV

For a known scalar function $g(Z)$, the IV estimator is

$$\hat{\alpha}_g^{IV} = \frac{\widehat{\mathrm{Cov}}(Y, g(Z))}{\widehat{\mathrm{Cov}}(D, g(Z))}.$$

Let

$$\varepsilon = Y - \mathrm{E}[Y] - \alpha_g^{IV}(D - \mathrm{E}[D]).$$

Then under finite moments,

$$\sqrt{N}\left(\hat{\alpha}_g^{IV} - \alpha_g^{IV}\right) \xrightarrow{d} N\left(0, \frac{\mathrm{E}\left[\varepsilon^2\{g(Z) - \mathrm{E}[g(Z)]\}^2\right]}{\mathrm{Cov}(D, g(Z))^2}\right).$$

Interpretation: The asymptotic variance is an IV variance formula for the LATE estimand. The appendix also explains why estimating $g(Z, \theta)$ in a first stage can affect the variance unless stronger conditional mean assumptions hold.

3.8 Examples in the paper

- **Draft lottery.** The instrument is the Vietnam-era draft lottery number. The LATE is for men whose military service status would change with lottery number.
- **Administrative screening.** The instrument is the assigned screening official. Exogeneity may be plausible, but monotonicity may fail if different officials use different criteria.
- **Randomized intention to treat.** The instrument is treatment assignment and actual treatment may differ because of noncompliance. The LATE is for assignment compliers.

Interpretation: These examples show that the assumptions are institutional, not just mathematical. LATE is credible only when the institutional mechanism makes exclusion and monotonicity plausible.

4 Identification and theoretical design

4.1 What identifies the LATE

- **Exogeneity/exclusion.** The instrument is independent of potential outcomes and potential treatment states.
- **First stage.** The instrument changes treatment probabilities.
- **Monotonicity.** Treatment status changes only in one direction when the instrument changes.
- **Finite moments.** Conditional means are well-defined.

Interpretation: The identified effect is local because it is tied to the individuals whose treatment is changed by the instrument.

4.2 What the paper does not identify

- It does not generally identify the average treatment effect for the whole population.
- It does not generally identify the average treatment effect on all treated individuals.
- It does not identify causal effects for never-takers or always-takers.
- It does not solve violations of exclusion or monotonicity.

Interpretation: IV is powerful, but the paper narrows its interpretation. The estimand is meaningful only after specifying the complier margin.

4.3 Why monotonicity is central

Condition 1 alone converts the reduced form into an average of signed treatment effects for movers. Monotonicity removes the negative-signed movers. With no defiers, the Wald estimand becomes an average over compliers.

Interpretation: Monotonicity is the bridge from an algebraic IV ratio to a causal parameter with a clear population interpretation.

4.4 Why the examples matter

The examples distinguish plausible and implausible uses of LATE:

- In draft lottery designs, monotonicity is plausible because a lower draft number should not reduce someone's probability of serving.
- In administrative screening, monotonicity may be implausible because different reviewers may favor different applicant types.
- In randomized encouragement or intention-to-treat designs, monotonicity is plausible when assignment can encourage take-up but not discourage it.

Interpretation: The paper tells researchers to justify the assumptions with design details, not with regressions alone.

4.5 What remains untestable

- Exclusion is generally untestable because it concerns potential outcomes under counterfactual instrument values.
- Monotonicity is generally untestable because it concerns individual-level counterfactual treatment responses.
- The identity of compliers is not directly observed for each individual.
- LATE may differ from other policy-relevant averages if treatment effects are heterogeneous.

Interpretation: The value of the theorem is not that it removes all assumptions. It clarifies exactly what assumptions deliver exactly what estimand.

5 Extracted theorem blocks

5.1 Extracted Block 1: Condition 1 - Existence of Instruments

Read:

- The condition requires joint independence of Z_i from potential outcomes and potential treatment decisions.
- It also requires instrument relevance: $P(w) = \mathrm{E}[D_i \mid Z_i = w]$ must vary with w.
- The exogeneity part is analogous to an exclusion restriction and is not directly testable.

Economic message: A valid instrument must both shift treatment and avoid directly shifting potential outcomes. Relevance can be checked empirically, but exclusion must be argued from the research design.

Condition 1 (Existence of Instruments): *Let Z be a random variable such that (i) for all $w \in \mathfrak{Z}$ the triple $(Y_i(0), Y_i(1), D_i(w))$ is jointly independent of Z_i, and, (ii) $P(w) = E[D_i | Z_i = w]$ is a nontrivial function of w.*

Condition 1: Existence of instruments, directly extracted from the paper.

5.2 Extracted Block 2: Condition 2 - Monotonicity

Read:

- Monotonicity requires the instrument to move treatment status in a common direction across all individuals.
- If $Z = w$ raises treatment probability relative to $Z = z$, then anyone who would take treatment at z must also take it at w.
- This rules out defiers: people whose treatment status changes opposite to the first stage.

Economic message: Monotonicity makes IV a clean average over compliers. Without it, positive and negative movers can cancel in the reduced form and destroy a causal interpretation.

Condition 2 (Monotonicity): *For all $z, w \in \mathfrak{Z}$, either $D_i(z) \geqslant D_i(w)$ for all i, or $D_i(z) \leqslant D_i(w)$ for all i.*

Condition 2: Monotonicity, directly extracted from the paper.

5.3 Extracted Block 3: Theorem 1 - Identification of the local average treatment effect

Read:

- Theorem 1 states the main LATE identification result.
- The estimand is the average treatment effect for people whose treatment changes when the instrument changes from w to z.
- The proof uses monotonicity to eliminate the opposite-sign mover term from the reduced-form decomposition.

- The Wald ratio is therefore a ratio of the reduced form to the first stage.

Economic message: The theorem is the foundation for interpreting binary IV estimates: the IV coefficient is a complier average causal effect.

THEOREM 1: *If Conditions* 1 *and* 2 *hold, then we can identify the following average treatment effect*:

$$\alpha_{z,w} = E[Y_i(1) - Y_i(0) | D_i(z) \neq D_i(w)]$$

from the joint distribution of Y, D, and Z, for all z and w in the support of Z such that $E[Y_i|Z_i = z]$ and $E[Y_i|Z_i = w]$ are finite, and $P(z) \neq P(w)$.

PROOF: Let Condition 2 be satisfied with $D_i(z) \geqslant D_i(w)$. Then $\Pr[D_i(z) - D_i(w) = -1] = 0$ which implies that the second term in (1) is equal to zero, and

$$E[Y_i|Z_i = z] - E[Y_i|Z_i = w]$$
$$= (P(z) - P(w)) \cdot E[Y_i(1) - Y_i(0) | D_i(z) - D_i(w) = 1].$$

Dividing both sides by $P(z) - P(w)$ shows that the local average treatment effect can be expressed in terms of moments of the joint distribution of (Y, D, Z). *Q.E.D.*

Theorem 1: LATE identification, directly extracted from the paper.

5.4 Extracted Block 4: Condition 3 - Monotone instrumenting functions

Read:

- Condition 3 governs scalar functions $g(Z)$ used with multi-valued or vector-valued instruments.
- The function $g(Z)$ must move monotonically with the treatment propensity $P(Z)$.
- The covariance $\mathrm{Cov}(D, g(Z))$ must be nonzero.

Economic message: This condition makes a multi-valued IV estimand interpretable as a nonnegative-weight average of LATEs instead of an opaque weighted average with possible sign reversals.

CONDITION 3: *$g(z)$ is a function from the support of Z to $\Re$ such that*
(*i*) *either for all $z, w \in \mathfrak{Z}$,* $\Pr(z) \leqslant P(w)$ *implies $g(z) \leqslant g(w)$, or, for all $z, w \in \mathfrak{Z}$, $P(z) \leqslant P(w)$ implies $g(z) \geqslant g(w)$;*
(*ii*) $\mathrm{Cov}(D, g(Z)) \neq 0$.

Condition 3: Monotone scalar instrumenting function, directly extracted from the paper.

5.5 Extracted Block 5: Theorem 2 - IV as a weighted average of LATEs

Read:

- Theorem 2 characterizes the IV estimand with a discrete multi-valued instrument.
- The IV ratio $\mathrm{Cov}(Y, g(Z))/\mathrm{Cov}(D, g(Z))$ equals a weighted average of adjacent local average treatment effects.
- The weights are nonnegative and sum to one if Conditions 1–3 hold.
- The weights depend on first-stage increments and the distribution of the instrument.

Economic message: Multi-valued IV coefficients are not automatically global effects. They are weighted averages of instrument-specific complier margins.

THEOREM 2: *Suppose that Conditions* 1, 2, *and* 3 *are satisfied. Let Z be a discrete random variable with support $\{z_0, z_1, \ldots, z_K\}$, ordered in such a way that if $l < m$ then $P(z_l) \leqslant P(z_m)$. Then, if $\operatorname{Cov}(D, g(Z)) \neq 0$, the IV estimator for the effect of D on Y using $g(Z)$ as an instrument estimates*

$$\alpha_g^{IV} = \operatorname{Cov}(Y, g(Z))/\operatorname{Cov}(D, g(Z)) = \sum_{k=1}^{K} \lambda_k \cdot \alpha_{z_k, z_{k-1}},$$

with weights

$$\lambda_k = \frac{(P(z_k) - P(z_{k-1})) \cdot \sum_{l=k}^{K} \pi_l \cdot (g(z_l) - E[g(Z)])}{\sum_{m=1}^{K} (P(z_m) - P(z_{m-1})) \cdot \sum_{l=m}^{K} \pi_l \cdot (g(z_l) - E[g(Z)])}, \tag{2}$$

where $\pi_k = \Pr(Z = z_k)$ and $\alpha_{z_k, z_{k-1}}$ is the local average treament effect $E[Y_i(1) - Y_i(0) | D_i(z_k) = 1, D_i(z_{k\text{-}1}) = 0]$. The weights λ_k are nonnegative and add up to one.

PROOF: See Appendix.

The second part of this section analyzes the asymptotic distribution of the IV

Theorem 2: IV as a weighted average of adjacent LATEs, directly extracted from the paper.

5.6 Extracted Block 6: Theorem 3 - Asymptotic variance of the IV estimator

Read:

- Theorem 3 gives the large-sample distribution of the IV estimator.
- The variance depends on the residual $\varepsilon = Y - \mathrm{E}[Y] - \alpha^{IV}(D - \mathrm{E}[D])$ and the centered instrumenting function $g(Z) - \mathrm{E}[g(Z)]$.
- In textbook homoskedastic settings, this collapses to the familiar IV variance expression.
- The appendix notes that estimating $g(Z, \theta)$ can affect the variance when stronger conditional error assumptions fail.

Economic message: The theorem connects the causal LATE interpretation to standard IV estimation and inference. It also warns that first-stage estimation can matter for standard errors.

Theorem 3: *Let* $(Y_i, D_i, Z_i)_{i=1}^N$ *be N independent, identically distributed random variables, and* $g(\cdot)$ *a function from the support of Z to* $\Re$ *such that* $\mathrm{Cov}(D, g(Z)) \neq 0$, *and let* $\hat{\alpha}_g^{IV}$ *be the instrumental variables estimator, given by*

$$\hat{\alpha}_g^{IV} = \widehat{\mathrm{Cov}}(Y, g(Z))/\widehat{\mathrm{Cov}}(D, g(Z)) = \frac{\sum_{i=1}^N g(Z_i)\cdot(Y_i - \overline{Y})}{\sum_{i=1}^N g(Z_i)\cdot(D_i - \overline{D})},$$

where $\overline{Y} = \sum_{i=1}^N Y_i/N$ *and* $\overline{D} = \sum_{i=1}^N D_i/N$. *Assume that all variances and covariances are finite. As N goes to infinity,*

$$\sqrt{N}\left(\hat{\alpha}_g^{IV} - \alpha_g^{IV}\right) \xrightarrow{d} \mathcal{N}\left(0, \frac{E\left[\varepsilon^2 \cdot \{g(Z) - E[g(Z)]\}^2\right]}{\mathrm{Cov}^2(D, g(Z))}\right),$$

with $\varepsilon = Y - E[Y] - \alpha_g^{IV}\cdot(D - E[D])$.

Proof: See Appendix.

In textbook discussions of instrumental variables estimation often the assumption $E[\varepsilon^2|Z=z] = \sigma^2$ is made. In that case the variance in Theorem 3 simplifies to the standard IV variance, $\sigma^2 \cdot \mathrm{Var}(g(Z))/\mathrm{Cov}^2(D, g(Z))$.

4. EXAMPLES

Theorem 3: Asymptotic distribution of the IV estimator, directly extracted from the paper.

5.7 Extracted Block 7: Three examples - Draft lottery, administrative screening, and randomized intention to treat

Read:

- The draft lottery example illustrates a plausible monotone instrument: lower lottery number raises draft risk.
- The administrative screening example shows why exogeneity is not enough; monotonicity can fail if reviewers use different criteria.
- The randomized intention-to-treat example gives the canonical compliance interpretation: assignment identifies the effect for compliers.

Economic message: The examples are the paper's applied guide. They show that LATE assumptions must be defended by the mechanics of the assignment rule.

Example 1 (Draft Lottery): Angrist (1990) uses the Vietnam-era draft lottery to estimate the effect of veteran status on earnings. The instrument is the draft lottery number, randomly assigned to date of birth and used to determine priority for military conscription. The average probability of serving in the military falls with lottery number. Condition 1 requires that potential earnings with and without military service be independent of the lottery number. This is a standard IV assumption which would be violated if, for example, lottery numbers are related to earnings through some variable other than veteran status. Condition 2 requires that someone who would serve in the military with lottery number k would also serve in the military with lottery number l less than k, which seems plausible. In the Angrist draft lottery application, $g(z)$ is an estimate of $P(z)$, and Condition 3 is therefore satisfied. The average effect of veteran status identified under Theorem 1 is for men who would have served with a low lottery number, but not with a high lottery number.

Example 2 (Administrative Screening):[5] Suppose applicants for a social program are screened by two officials. The two officials are likely to have different admission rates, even if the stated admission criteria are identical. Since the identity of the official is probably immaterial to the response, it seems plausible that Condition 1 is satisfied. The instrument is binary so Condition 3 is trivially satisfied. However, Condition 2 requires that if official A accepts applicants with probability $P(0)$, and official B accepts people with probability $P(1) > P(0)$, official B must accept *any* applicant who would have been accepted by official A. This is unlikely to hold if admission is based on a number of criteria. Therefore, in this example we *cannot* use Theorem 1 to identify a local average treatment effect nonparametrically despite the presence of an instrument satisfying Condition 1.

Example 3 (Randomization of Intention to Treat):[6] Let the instrument be an indicator for assignment to treatment group in a randomized trial. The actual treatment indicator, D, may differ from the instrument Z because some individuals may not comply with their assignment. Condition 1 requires that the two counterfactual outcomes, say health status if treated and health status if untreated, are independent of the original assignment. Condition 2 requires that anyone who would take the treatment if assigned to the control group would also take the treatment if assigned to the treatment group. This seems plausible if noncompliance is the result of a decision by patients. The instrument is binary so Condition 3 is satisfied. The treatment effect identified here is the average treatment effect for those who always comply with their assignment.

Examples section: institutional interpretation of LATE assumptions, directly extracted from the paper.

6 Short summary

- Imbens and Angrist define what IV estimates when treatment effects differ across people.
- The framework uses potential outcomes $Y_i(0), Y_i(1)$ and potential treatment states $D_i(z)$.
- A valid instrument must be independent of potential outcomes and potential treatment decisions, and it must shift treatment probabilities.
- Exogeneity alone is not enough for point identification of a meaningful average effect.
- Monotonicity rules out defiers and makes the reduced form proportional to the average causal effect for compliers.
- With a binary instrument, the Wald ratio identifies the local average treatment effect for compliers.
- With a discrete multi-valued instrument, IV identifies a nonnegative-weight average of adjacent LATEs under monotone $g(Z)$.
- The target population is determined by the instrument, so different instruments can identify different treatment effects.
- The paper gives asymptotic theory for IV estimators of LATE.
- The examples show that institutional knowledge is essential for judging exclusion and monotonicity.
- The paper does not identify ATE, ATT, or treatment effects for always-takers and never-takers without stronger assumptions.
- The lasting lesson is that IV estimates are causal only after specifying the relevant complier population and assumptions.

7 Conclusion

7.1 Core conclusion

Imbens and Angrist show that instrumental variables can identify a well-defined causal parameter without constant treatment effects or parametric selection models. The parameter is the local average treatment effect: the average effect for individuals whose treatment status is changed by the instrument. This insight reconciles IV with heterogeneous treatment effects by changing the estimand from a global average to a local complier average.

7.2 Economic intuition

The intuition is simple. An instrument changes outcomes only through people whose treatment status changes. Always-takers are treated regardless of the instrument, and never-takers are untreated regardless of the instrument. Their outcomes may help determine conditional means, but they do not contribute to the causal change induced by the instrument. If there are no defiers, the reduced-form effect of the instrument on the outcome is exactly the first-stage effect times the average treatment effect for compliers.

7.3 What researchers should report

The paper implies that applied researchers should report more than a first stage and a second stage:

- Describe the institutional mechanism that makes the instrument exogenous.
- Explain why treatment take-up should be monotone in the instrument.

- Identify the likely complier group.
- Avoid interpreting IV as an ATE unless additional assumptions justify that interpretation.
- For multi-valued instruments, explain how the instrumenting function weights different margins.

7.4 Why the paper changed empirical practice

Before this paper, IV was often interpreted as a way to estimate "the" treatment effect after correcting endogeneity. The LATE framework clarified that IV estimates a particular weighted average of heterogeneous effects. This shifted empirical practice toward design-based interpretation: the credibility and meaning of an IV estimate come from the assignment mechanism and the population it moves.

7.5 Limits and cautions

The LATE estimand is precise but local. It may not answer every policy question. A draft-lottery LATE is about people induced into military service by draft risk, not about volunteers or people who would never serve. An intention-to-treat LATE is about compliers, not always-takers or never-takers. Monotonicity and exclusion are not automatically true; they must be defended. The paper's contribution is to make these limits explicit rather than hiding them inside a single IV coefficient.

7.6 Takeaway

The enduring message is that IV is not only an estimation technique; it is a causal design with a specific target population. Under exclusion, relevance, and monotonicity, the IV coefficient is interpretable as the average causal effect for compliers. That interpretation is powerful, but it is local by construction.